

PHILIPPINE NATIONAL STANDARD

PNS/BAFS 142:2018
ICS 67.060

Code of hygienic practice for processing and handling of corn grits



BUREAU OF AGRICULTURE AND FISHERIES STANDARDS

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Foreword

The Philippine National Standard (PNS) Code of hygienic practice for processing and handling of corn grits was developed in 2015 and reviewed together with other corn-related standards in 2017 by the Bureau of Agriculture and Fisheries Standards (BAFS) to check if its provisions are still relevant and effective to the current regulatory and market needs. It has been revised by the Technical Working Group (TWG) for the review of various PNS related to corn as per Department of Agriculture Special Order Nos. 301 and 814 Series of 2017. This Standard has been approved by the Secretary of the Department of Agriculture in 2018.

The provisions of this document are supplemental to and should be used in conjunction with PNS for Corn (maize) grits – Grading and classification (PNS/BAFS 15).

This PNS/BAFS 142:2018 cancels and replaces PNS/BAFS 142:2015 which has been technically revised.

This edition includes the following significant changes compared to the previous definition:

- updating of referred standards and inclusion of other normative PNS; and
- removal of provisions pertaining to Good Manufacturing Practices (GMP) requirements, product recall, and sanctions.

This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2.

Introduction

Corn or maize is one of the most popular cereals in the world and forms the staple food in many countries including the Philippines. According to the study made by World Food Program (2012), white corn processed into grits is the staple food of 20% of the Filipino population primarily in the Visayas and Mindanao area.

Health benefits of corn grits have been undeniable. Corn grits is a low-fat complex carbohydrate with low glycemic index that deserves a regular place on our healthy diet and highly recommended for people with diabetes. Corn grits is also popularly mixed with rice called “rice composite” which is more nutritious and filling than rice alone.

The creation of the code of hygienic practice for processing and handling of corn grits is important to ensure the safety and quality of the produced corn grits. It will help promote corn grits as alternative or complementary staple to rice and provide awareness to consumers of its health and wellness benefits.

With the Philippine National Standard (PNS) Corn (maize) grits – Grading and classification (PNS/BAFS 15) already concerned with the classification of corn grits based on their physical characteristics and current practices in existing sectors, this Code aims to provide guidance for the processing, handling, and storage of quality corn grits in order to prevent the risk of contamination and infestation of the said product with hazards like aflatoxin, heavy metals, and pesticide residues.

1 Scope

This Code of Hygienic Practice (COHP) applies to the overall production of corn grits derived mainly from flint corn grains through proper harvesting, drying, shelling, milling, packaging, and storage.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

PNS/BAFS 15, *Corn (maize) grits – Grading and classification*

PNS/BAFS 193, *Code of Good Warehousing Practices for bagged grains*

PNS/BAFS/PAES 251, *Agricultural Machinery - Corn Mill - Specifications*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1

corn grits

milled corn grains where the outer covering (pericarp) and germ (embryo) have been removed leaving only the endosperm that passed through different sieve sizes

3.2

Good Manufacturing Practices (GMP)

quality assurance system aimed at ensuring that products are consistently manufactured, packed, repacked or held to quality standards appropriate for the intended use. It is thus concerned with both manufacturing and quality control procedure

4 Primary production

4.1 Raw materials

The principles and guidelines supplement those contained in PNS Grains – Grading and classification - Corn (PNS/BAFS 10).

4.2 Environmental hygiene

The farm environment, farm structure and facility maintenance for corn should conform with PNS Good Agricultural Practices (GAP) for corn (PNS/BAFS 20).

4.3 Hygienic production of corn grits

4.3.1 Pre-processing

It is recommended that the raw materials covered in this section be prepared and handled in accordance with the appropriate sections of the PNS Code of practice for the prevention and reduction of aflatoxin contamination in corn (PNS/BAFS 27) and PNS/BAFS 20.

4.3.1.1 Harvesting

Harvesting practices should conform with PNS/BAFS 20 and PNS/BAFS 27 in order to prevent damage and contamination of corn ears to be processed to corn grits.

4.3.1.2 Dehusking

The husk should be properly removed after harvest prior to drying or shelling depending on the moisture content of the corn grains.

4.3.1.3 Drying

4.3.1.3.1 First stage drying:

Sorting should be done prior to and after drying to remove poor quality corn ears.

Matured corn ears should be pre-dried using mechanical and/or sun drying within 24 hours after harvest to reduce the moisture content (MC) to not more than 20 % to reduce grain breakage during shelling.

4.3.1.3.2 Second stage drying:

Shelled corn grains shall be dried to not more than 14 % MC to avoid or minimize grain deterioration from mold and insect infestation using the appropriate methods of drying.

4.3.1.4 Shelling

Corn ears should be shelled after first stage drying. Sheller should be cleaned and dried properly before use.

4.3.2 Storage and transport of corn grains

Storage and transport of bagged corn grains shall be in accordance with the PNS Code of Good Warehousing Practices (GWP) for Bagged Grains (PNS/BAFS 193).

4.3.3 Processing

The corn mill to be used for milling shall conform with the PNS Agricultural Machinery - Corn Mill - Specifications (PNS/BAFS/PAES 251).

4.4 General hygienic practice

4.4.1 Contaminants control

The level of aflatoxin, heavy metals, live insects, and pesticide residues shall conform with the appropriate guidelines specified in the PNS/BAFS 15.

4.4.2 Packaging

The packaging materials to be used shall be suitable and compliant with PNS/BAFS 15.

4.4.3 Storage

4.4.3.1 The storage area for bagged corn grits should be designed and constructed to:

- prevent wetting of corn grits and maintain the recommended moisture content to a maximum of 14 % MC during storage to ensure product quality;
- minimize the entry of insects, birds, rodents, and other animals using appropriate measures; and
- provide proper ventilation for the stored bagged corn grits.

4.4.3.2 The processors should conduct regular monitoring of corn grits quality during storage and prior to distribution. No corn grits should be released to the market containing more than 20 µg/kg (ppb) aflatoxin level.

4.4.4 Equipment and utensils

Equipment, containers and utensils coming into contact with the product should be designed and constructed using materials that are easily cleaned, disinfected, and regularly maintained to avoid contamination of corn grits.

4.5 Documentation and record keeping

Corn mill operators should have updated records of corn grits production to include procurement, processing, storage and distribution as integral part of the food safety control system. All records should be archived for at least 12 months.

5 Establishment: Design and facilities

5.1 Plant construction and layout

5.1.1 Site and premises of the corn mill

5.1.1.1 The site including its facilities should follow environmental, zoning, and other relevant requirements or regulations set by concerned competent authorities. Preferably, it should be built in agro-industrial zones and away from residential areas.

5.1.1.2 Potential sources of contamination have to be considered in the location of the corn mill. In particular, corn mills should not be in:

- areas with industrial and other activities which pose a serious threat of contaminating corn grits;
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- flood prone/vulnerable areas;
- areas prone to storage pest infestation;
- areas where wastes (either solid or liquid) cannot be removed effectively.

5.1.1.3 Mill owners/operators should secure immediate perimeter with adequate fences or gate to prevent any unwanted intrusion whether by humans and/or animals.

5.1.2 Building and plant facilities

It is recommended that the provision of this section be referred to the appropriate section of the Department of Health Administrative Order No. 153 s. 2004 Revised Guidelines on Current Good Manufacturing Practice in Manufacturing, Packing, Repacking, or Holding Food or its latest issuance and National Food Authority Revised Rules and Regulations on Grains Business.

6 End product specification

Corn grits shall conform with the quality grade requirements as stated in PNS/BAFS 15.

7 Product information

7.1 Labeling

Labeling of corn grits shall be in accordance with the provisions of PNS/BAFS 15.

7.2 Traceability

Each product package should be traceable to the source (miller/processor).

Miller/processor should maintain retained sample for reference.

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PNS/BAFS 10, *Grains – Grading and classification - Corn*

PNS/BAFS 20, *Good Agricultural Practices (GAP) for corn*

PNS/BAFS 27, *Code of practice for the prevention and reduction of aflatoxin contamination in corn*

Republic Act No. 10611, *An act to strengthen the food safety regulatory system in the country to protect consumer health and facilitate market access of local foods and food products, and for other purposes*

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